AlignΞd

A Smart Neoprene Knee Brace to Aid Beginners in Confident Squat Training

Detailed Design, Manufacture, Assembly and Production Report

Group 12
Zayn Muntazir, Yousuf Shahabuddin, James Xu, William Tu

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1 Executive Summary

Align∃d is a battery powered, hand operated electromechanical knee brace that aims to help gym beginners squat with safe, controlled form. It can be used by all and in various situations, such as body weight squats, free weight squats, by ones self, with a trainer and more. This is the answer to issues that were identified after 2 months of primary and secondary research, including beginner social pressures, incorrect form and fear of injury.

The product is a discrete neoprene sleeve that has a detachable electronics pod containing and IMU, alongside other components and circuits, to track the angle of the knee. A linear resonant actuator (LRA) gives a 'sharp tap' to the inside of the knee when the IMU detects the knee valgus angle going above a dangerous threshold, one of the largest factors to injury during this exercise according to an interview with a personal trainer. A vibration motors helps with squat tempo, and buzzing when at good depth. Alongside this, there is a mechanical BOA-dial to tighten the brace, providing a 'one-size' compression. This is all powered by a 300mAh cell based on the power and time requirements of the brace.

Under the brand of Hyperice, *Align∃d* is engineered for mass production, being optimised from design for assembly, manufacture and disassembly point of views. This is shown in examples that the two shot moulded outer case that is closed using ultrasonic welding, providing a satisfying finish. Based on the bill of materials and manufacturing, the product aims to be sold at a retail price of £98, a defensible price based on the users, specs and business case.

2 Concept Development

2.1 The Problem

As humans, we must look after and maintain our body, since we only get one. Based on two months of research, we found that there were a large number of people who wanted to start going to the gym or general fitness, but were too worried to start for various reasons, such as:

- Fear of incorrect form
- Fear of judgment
- Fear of injury
- Unsure where or how to start
- Lack of time and money

This was the beginning of *Align∃d*. Beginners attempting free-weighted squats without professional assistance are at significant risk of injury; either compounded over time due to poor form or a major injury event. This mainly comes from one factor: leg positioning, specifically the knee valgus, as stated by a personal trainer during one of our interviews.

2.2 Current Market

The current gym market is geared towards professionals and veterans (those who have been going to the gym for a while), and anything that is usable by beginners is either behind a massive paywall or would be time consuming. One might be able to find some beginner products, but they are usually very low quality and gimmicky.

As well as this, products usually fit into three broad categories, each with their own limitations:
Analogue Equipment
Wearable Fitness Trackers
App-based Tracking and Analysis
This leaves a clear gap in the market that *Align∃d* aims to fill. A discrete, wearable tracker that provides real-time feedback to correct form.

2.3 Final Concept

After various rounds of ideation and analysis, the final concept was selected using a weighted decision matrix against five criteria: Beginner Accessibility (x1.0), Impact on Movement Quality (×1.25), User Confidence (×1.0), Feasibility (×1.1), and Fit with Brief (×1.0). The factors were weighted based on importance for the use case and user group.

This, alongside other states of analysis and synthesis, led us to our final idea: a smart neoprene knee brace with an in-built, real-time, haptic response system to correct form. Enclosed in a small detachable case, the electronics pod contains the microcontroller, IMU and batteries, with the haptic actuators (LRA and ERM vibration motor). On the side of the brace, there is a mechanical tightening mechanism to allow for a 'one size fits all' brace.

3 Requirements and Specifications

The requirements and specifications were developed from the product aims, user needs, safety considerations, and technical constraints of the electromechanical knee brace system.

Table 1: Product requirements and specifications for the smart knee brace system

No.	Requirement	Specification	Verification Method
R1	Knee alignment feedback	The system shall detect excessive knee valgus during squatting and provide feedback when the knee deviates beyond the acceptable threshold.	Controlled squat test with different knee positions.
R2	Real-time response	Feedback shall be provided within 300 ms of detecting incorrect knee movement.	Measure the delay between sensor input and feedback output.
R3	Detachable electronics pod	The electronics pod shall be easily removable without tools while remaining secure during normal squat movement.	Attachment, removal, and repeated squat testing.
R4	Comfort	The brace shall not restrict normal squat movement or cause discomfort during exercise sessions.	User testing and comfort questionnaire.
R5	Battery life	The system shall operate continuously for at least 45 min on one charge.	Continuous run-time test under normal use.
R6	Safety	All exposed electronics shall be enclosed, with no sharp edges or conductive contacts touching the user.	Visual inspection and handling test.

4 Materials and Manufacturing

This section details the material choice, manufacturing process and design rationale of each component of the product. All decisions are considered in the context of a mass-production wearable sports performance device with a target user group of beginners, taking into account: cost, reliability, skin safety, ease of use and professionalism.

4.1 Knee Brace

Material

The brace will be made from SCR neoprene (jersey-lined on both faces), an industry standard for sports compression braces, as used by Bauerfeind [1]. It provides the needed compression to support the joint, alongside thermal retention and a foam structure that resists sweat absorption. [11] The jersey lining provides a soft skin contact on the inner side, as well as a stable-bonding substrate on the outer side.

Manufacturing

The sleeve is formed from a flat-cut SCR neoprene blank, contoured to the shape of the knee using a curved seam pattern. The seem is bonded and stitched: neoprene-compatible contact cement and a flat-lock stitch using an industrial flat-lock machine.

Finishing

The upper and lower edges of the brace are finished with a jersey-covered neoprene binding tape in the brand colours that would be sewn and bonded to the raw edge. This helps prevent rolling and provides a professional finish to the product.

The patella cutout is edged with a TPU ring that eliminates pressure on the patella during deep flexion and helps the user know the correct direction of wearing. It is injection moulded as a separate component and bonded using a compatible urethane adhesive.

4.2 Tightening Mechanism

Since *AlignΞd* is using a BOA Fit Mechanism [2], which is propriety technology, there is a choice of either becoming a brand partner as over 300 brands have done [10], or making a similar dial-tightening system. Since BOA Fit are leaders in this field, there is great benefit of utilising their technology, which might strengthen the brand image of *AlignΞd* further. This therefore means that the manufacturing of the system is outsourced, but what is done in house is integrating this subsystem into the greater product.

4.3 Electronics Pod

The pod is made detachable by having 2 parts: the main pod and dock plate. This allows the user to wash the knee brace when needed, as well as allowing

for simple charging.

Housing Pod

Since the PCB and batteries need to be encased within the pod, the manufacturing must be split into an upper and lower shell. This also opens the opportunity to vary the material of the 2 halves for whatever beneficial.

Materials

The **lower shell** will be made from PA6-GF30 (Polyamide 6, 30% short glass-fibre reinforced). The glass reinforcement raises the stiffness significantly, resisting any creep from the magnetic clamping force.

The **upper shell** will be made of two materials: PA6-GF30 for the inside, and TPU 95A for the outer overmould. This allows the top layer to remain strong to protect the PCB, while also providing a premium feel to the surface of the pod for the user experience. Due to the TPU's flexibility, it also allows for the living hinge feature on the USB-C port to protect from sweat and moisture.

Manufacturing

The **lower shell** will be produced by conventional injection moulding in a single-cavity hardened P20 steel tool. All features (pogo bores, alignment pins, battery recess) will be included in the mould and draft angles of 1.5° on all vertical walls will allow for a clean ejection.

Due to the dual material approach, the upper shell will be produced as a two-shot moulded component in a dedicated 2K tool. In the first shot, the PA6-GF30 outlines the inner shape, weld rim geometry and button apertures. The second shot containing the TPU is then injected over the top face, forming the exterior skin and the living hinge port cover.

After all of the components have been placed into the lower shell (PCB and batteries), the pod must be closed. This will be done with ultrasonic welding, a horn vibrating at $40kHz$) welds the two parts. This results in a seamless transition between the two parts, giving a better user experience as well as contributing to IP54 compliance.

Dock Plate

Material

The dock plate will be made from PA6-GF30 as it needs to be stiff and creep-resistant under the constant magnetic loading and unloading. As well as this, matching the material between the lower shell and dock plate ensures identical thermal expansion coefficients, meaning they should last longer together.

Manufacturing

It will be injection moulded in a straight-pull two-plate tool as the geometry is very simple. All features, such as the magnet pockets and alignment pin holes, are included in the mould and parallel to the mould pull direction. There will be draft angles of 1.5° on all walls.

4.4 Complete Product

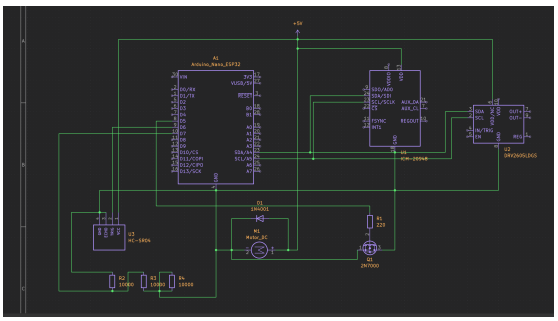
The assembly of the product involves using N25 magnets in the pod body and the base plate on the brace. The magnets are orientated with opposing poles so that the user cannot place the pod on incorrectly, and the magnets have a pulling force of 6kg, ensuring the pod stays attached securely to the brace. Asymmetrical alignment pegs are used in addition to prevent user error.

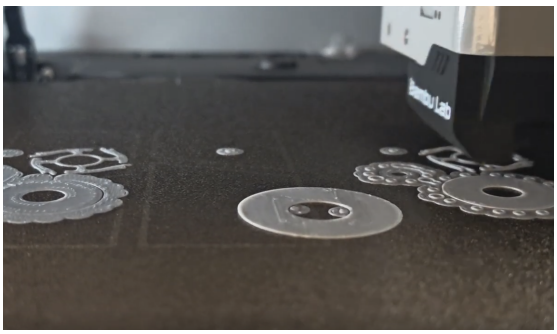
The BOA system would be stitched into the knee brace, through holes left for sewing, as this makes the system removable should there be any faults or wear and tear that would constitute a necessary replacement of the part(s).

5 Prototyping Plans

5.1 Electronics

The circuit schematic was created using KiCad, as it allowed for us to model our components and eventually produce a PCB if needed. Initially, we modelled using a breadboard, however opted to use a perforated board due later to its simplicity and testing flexibility, whilst being relatively compact.





5.2 Mechanism

Rather than having straps for tightening, we opted for BOA fit system- they are internally complex, and so we based our designs off an existing one we found to understand how the system functions. The wire was purchased, as well as standard parts such as screws and bolts. We chose the BOA fit mechanism as it is a proven simple-to-operate mechanism, widely used within sports gear, as an adapted recreation of the BOA fit system [13].

5.3 Manufacturing

The majority of our components were printed using a Bambu Lab X1 Carbon with PLA filament, due to the relatively low-cost and fast rapid nature of 3D printing, allowing us to iterate designs and test our ideas quickly.

As our project was not investigating the biomechanics of knee braces, we opted to test the shape using foam and cardboard, before transitioning to an elasticated knee brace to build on top of. We began with deciding what components were needed based on our initial designs. To secure the tightening chords to the brace for tightening, we 3D printed hooks.

6 Power and Load Calculations

In order to have a successful product, one must optimise the battery choice based on both the electronics and the desired operation time. To collect the necessary data to use in calculations, all the sensors and actuators were tested in the idle and active states, and the current and voltage was recorded using a multimeter. A snapshot of the data follows:

Table 2: Measured current consumption of system components

Measurement	**LRA Idle** (mA)	**LRA Active** (mA)	**Vibration Motor** (mA)	**Ultrasonic Sensor** (mA)	**ESP32** (mA)	**IMU** (mA)
Current 1	0.58	1.30	30.4	1.71	20.2	5.97
Current 2	0.59	2.40	29.7	1.70	20.7	6.04
Current 3	0.58	1.79	30.3	1.69	20.5	6.05
Average Current	0.58	1.83	30.13	1.70	20.47	6.02

Comparing the experimental data with the datasheets for each of the components, the experimental data was within a suitable range of the actual data for each of the components. Based on this, it was decided that it would be best to use the provided data from each company as that would have been through more rigorous testing.

6.1 Idle Current Draw

To begin the calculations, the idle current, the current while the device is on but no active event is being triggered, must be measured, as in the scope of this product, this is the more common of the modes.

6.2 Peak Current

A valgus event fires the LRA through the DRV2605L's overdrive phase [9]. The largest transient:

$$I_{\text{peak,LRA}} = 24 + 0.76 + 150 = 174.76\,\text{mA}\ (576.7\,\text{mW})$$

A depth event runs the ERM at 80% PWM ($3.3\times0.8 = 2.64\,\text{V}$ effective, $85 \times \frac{2.64}{3.0} \approx 74.8\,\text{mA}$):

$$I_{\text{peak,ERM}} = 24 + 0.76 + 0.65 + 74.8 = 100.21\,\text{mA}$$

Independent firmware cooldowns (LRA 600 ms, ERM 1200 ms) prevent the two from coinciding by design.

6.3 Duty Cycle and Average Current

Over a 60-minute beginner session (5 sets, brief actuator events), the time-weighted average is overwhelmingly idle-dominated:

$$I_{\text{avg}} = 23.89 + 0.36 + 0.63 \approx 24.88\,\text{mA}$$

The actuators contribute $< 4\%$ of average current, the key design insight is that MCU idle current dominates, so a future modem-sleep between IMU samples would yield the largest battery gain.[3], [5],[7], [12]

7 Design for Manufacture

The following section reflects on the design and manufacturing decisions that allow *AlignΞd* to reduce cost, complexity and streamline the process.

Part count reduction: Arguably the most efficient decision as it pertains to DFM, as well as DFA, is combining parts where possible, and removing anything unnecessary. This can be seen in the pogo pin housings, magnet pockets, PCB rails and alignment bosses, all integral features, collapsed into a single part through the choice of manufacturing process, making manufacturing simpler.

Mouldability All features are parallel to the direction of pull out of the mould, with draft angles of 1.5° and uniform walls with chamfered transitions avoids any sink or warp and is overall materially cheaper than a side-action tool.

Two-shot upper shell PA6-GF30 and TPU bond via hydrogen bonding [3], removing some potential manufacturing complications and adding more use.

PCB

Removes majority of soldering work with pick and place pieces, only hand soldering is four wires to the pogo pins which is very simple.

8 Design for Assembly

AlignΞd was analysed and optimised for DFA, to minimise assembly time, reduce the chance of human error and incorrect assembly and to allow single-operator production with limited specialist tooling.

Positive stops: Hard pockets, PCB rails and the general shape mean that parts can't be over-inserted or misplaced.

Self location: PCB rails make placement clear in all three-axis and the magnets self-register the positioning of the pod.

Error Proofing: Aspects like reverse polarity magnets and colour coded wires make it hard to make errors in placement or assembly.

Modularity: The three part architecture of the sleeve, mechanism and pod is a DFA is a strength of the product as it allows for each subsystem to be built and tested separately, making it possible to test along the way and pinpoint any errors.

9 Design for Disassembly and End of Life

Disassembly was designed at three levels per ISO 8887 [8]: user level, service level and end of life, with time and material separation targets set clearly.

User level: A two finger pull against against the magnets detaches the pod for charging and washing, with reattachment made very simple by the magnet and alignment pins. This level of disassembly is what a user needs for all wanted functions: use during exercise, charging easily and washing safely.

Service level: The pod is sealed and can be replaced as a whole unit under warranty, but it can be opened by a technician in a non-destructive manner to access the PCB and electronics. The sleeve, dock plate and Boa dial are each independently replaceable without needing to discard the electronics, supporting a lower cost of ownership.

End of Life material separation: Polymers have ISO 11649 identification marks. Shells separately cleanly at the weld joint. The lithium cell removed without tools, as stated by EU 2023/1542 [6]. PCB and ERMs will go to WEEE. Magnets can be recovered by IPA-softened UV adhesive. Neoprene and Dyneema go to textile/fibre streams.

Table 3: End-of-life material separation summary (whole product).

Component	**Material**	**Separation**	**Route**
Pod shells	PA6-GF30 (+TPU skin)	Cut weld; peel/shred TPU, hydrocyclone density split	PA6 / TPU recyclate
PCB assembly	FR4, Cu, Au, Ni, Si	Lifts out after weld cut	WEEE (Au/Cu recovery)
Battery	LiPo	Keyed JST, no adhesive	EU 2023/1542 battery recycling
Magnets	NdFeB (Ni-Cu-Ni)	IPA softens UV adhesive	Rare-earth recovery
Dock + sub-plate	PA6-GF30, FR4/ENIG	Seam-rip bar-tack; blade off FR4	PA6 recyclate / WEEE
Sleeve + binding	SCR neoprene, jersey	FPC peels at 60°C	Textile / neoprene stream
Boa dial	PA6-GF30, 316 steel, UHMWPE	Unpick stitch, IPA bond, mech. disassembly	PA6 / steel / fibre

10 Product Assembly (Details and Analysis)

10.1 Electronics Pod Sub-Assembly

Table 4: Electronics pod sub-assembly sequence

Assembly step	**Time**
Dispense UV adhesive into four magnet slots. Insert N52 magnets, polarity verified against a reference board. Cure at 405 nm for 10 s per pocket.	45 s
Press four pogo pins into Ø1.1 mm bores from the exterior until flanges seat flush. Apply CA adhesive around each barrel inside.	30 s
Place PCB inverted on a jig; solder pogo-pin barrels to the pads. Inspect all joints. (Manual tool only.)	40 s
Slide PCB into the lower-shell guide rails. The USB-C aperture self-checks orientation, preventing mis-installation.	20 s
Place LiPo battery into the moulded recess; connect the keyed JST-PH connector to the PCB header.	20 s
Fit the upper lid and ultrasonic-weld for 0.8 s. An integrated pull-force sensor verifies weld quality.	25 s
Automated end-of-line test: power-on, IMU self-check, pogo continuity (×4), LRA and ERM actuation + current, LED, charge-circuit validation.	60-90 s
Total active assembly time	**≈ 3 min**

10.2 Dock Plate Sub-Assembly

Table 5: Dock plate sub-assembly sequence

Step	Expected time
Solder wire tails to the underside of all four ENIG pads on the FR4 sub-plate, done flat before insertion for full joint access.	60 s
Press FR4 sub-plate into the dock-plate recess until flush. Apply UV adhesive around the perimeter; cure at 405 nm.	20 s
Route wire tails through the base slot. Pre-crimp JST connectors onto the tail ends for FPC connection.	20 s
Probe continuity across all four pogo pads. Reject any sub-assembly below 30 mΩ contact resistance.	15 s
Total active assembly time	**≈2 min**

10.3 Knee Sleeve Sub-Assembly

Table 6: Neoprene sleeve garment sub-assembly sequence

Step	Expected time
Die-cut neoprene blank from sheet. All apertures (patella cutout, dock plate adhesive face, cord clip post holes, FPC routing slot) cut in a single press operation.	60 s
Apply neoprene contact cement to posterior seam faces. Press-form and flatlock-stitch posterior seam on industrial sewing machine to produce anatomically contoured tubular body.	90 s
Bond and sew teal jersey-lined binding tape to upper and lower edges. Dispense silicone grip strip bead at both inner edges by automated needle applicator.	60 s
Bond moulded TPU patella ring to cutout edge with urethane adhesive. Bar-tack at four points for mechanical retention.	45 s
Heat-stake six cord guide clips through pre-punched anterior holes, three bilateral pairs at 40 mm pitch. Apply backing plate to inner face of each.	60 s
Stitch two LSR silicone actuator pockets to inner face at lateral knee position, open face toward skin.	30 s
Cure silicone grip strips and any remaining adhesives at 60 °C in batch oven. *(Parallel operation across multiple sleeves.)*	30 s
Total active assembly time	**≈6 min + oven cure (parallel)**

10.4 Product Assembly

Table 7: Whole-product integration assembly sequence

Step	Expected time
Peel adhesive backing from FPC and press-laminate to inner sleeve face along pre-marked routing path from dock plate face to actuator pocket positions.	60 s
Prepare dock plate bonding face and sleeve lateral panel: abrade lightly, wipe with IPA. Apply structural PU adhesive (both faces), allow flash-off time per manufacturer specification.	90 s
Press dock plate onto prepared sleeve face. Hold under light compression during initial cure. Route JST wire tails through base slot and click to FPC termination.	30 s
Position BOA dial housing plate at pre-marked anterior location. Sew four corners on industrial sewing machine. Apply neoprene contact cement to full plate perimeter and hold.	120 s
Thread pre-swaged Dyneema cord through PTFE grommets and guide clip sequence: upper bilateral pair → cross → mid bilateral pair → cross → lower bilateral pair. Seat swaged ferrule in lower anchor pockets.	120 s
Press LRA into designated LSR pocket. Press ERM into designated LSR pocket. Connect colour-coded JST Pico connectors: LRA white, ERM black. Audible click confirms seating.	45 s
Dispense flowable LSR around each seated actuator using dispensing needle. Transfer to conveyor oven at 80 °C for 30 min. *(Parallel across multiple units; zero time on critical path.)*	30 s
System functional test with pod magnetically docked: pogo continuity, LRA and ERM current draw, IMU self-test, BLE advertisement, LED state. Calibrated vibration sensor placed on LSR patch surface confirms perceptible output at skin-contact face.	90-120 s
Dial function check: rotate through full tightening range, verify ratchet click count, confirm full cord slack on release-button press.	30 s
Pack with pod docked in sleeve. Insert quick-start card. Seal retail carton.	45 s
Total active assembly time	**≈18 min**

Whole product total: Pod (≈4 min) + Dock plate (≈2 min) + Sleeve (≈6 min) + Integration (≈18 min) = ≈30 min active per unit.

11 Estimated Cost

To estimate the cost per unit, we used a bottom-up cost model for a production run of 10,000 units, which is consistent with the business case for mass market sports devices. It must be noted that these numbers a predominantly estimates and are based on various assumptions:

11.1 Assumptions

- Components will be priced cheaper when bought in bulk.
- Operating costs estimated excluding purchase of machinery and tooling.

Table 8: Indicative unit cost at volume (≈50k/yr).

Item	Est. cost	Item	Est. cost
PCB assembly (ESP32-S3 module, IMU, DRV2605L, passives, board)	£9.20	Neoprene sleeve (cut & sew)	£4.50
LiPo cell 300 mAh	£1.80	Boa dial + cord (OEM licensed)	£3.50
LRA + ERM actuators	£2.30	FPC + JST connectors	£0.90
Lower shell + 2K upper lid	£1.10	LSR pockets + potting	£0.80
Dock plate + FR4/ENIG sub-plate	£0.55	Patella ring, clips, grip strips	£0.65
Pogo pins ×4 + N52 magnets ×4	£1.00	Adhesives/consumables	£0.40
Retail packaging + quick-start card	£1.20	**BOM subtotal**	**≈£28.4**
Assembly labour (≈30 min, mostly parallel/automated)	£3.00	Test, QA, overhead, logistics	£6.0
Estimated factory cost		**≈£37-40 / unit**	

12 Branding CMF Strategy, Development and Implementations

12.1 Brand Selection

In the brand design stage, the team referred to several brands from neighbouring markets, including Gymshark, WHOOP, Bauerfeind and Hyperice. These brands were selected because they represent the different directions that smart knee pads may develop: fitness culture, wearable performance tracking, medical assistance and sports recovery technology.

Gymshark can be used as a reference for fitness culture, self-confidence and healthy lifestyle, but its brand language mainly revolves around clothing and cannot fully convey the sensor feedback and electronic guidance system in our products.

WHOOP provides a powerful reference for performance tracking and data-driven training of wearable devices. Its concise digital image and attention to user insights are relevant, but the brand is more related to passive monitoring than real-time body movement correction.

The reason why we considered **Bauerfeind** is that it has a good reputation in knee support and orthopaedic products. However, its clinical and medical properties may make people feel that the product is an auxiliary device for post-injury rehabilitation, which conflicts with our goal of positioning it as a daily training aid for beginners in the gym.

Hyperice was selected as the final brand reference for the following reasons:

- **Sports-tech positioning:** Hyperice combines sports performance, recovery technology and user-friendly hardware to make it highly compatible with intelligent sports feedback products.
- **Physical feedback relevance:** Hyperice products often use tactile and physical interaction methods, which echos the way we use LRA and vibration motor feedback in squat training.
- **Non-clinical reliability:** The brand has a sense of technology and is trustworthy, but it does not show medical equipment, which helps to avoid users from having the negative impression of injury or relying on assistance.
- **Beginner-friendly confidence:** Hyperice's high-end and popular brand image provides support for users who want to get guidance, enhance confidence and improve movement perception in training.
- **CMF and communication influence:** Its concise, minimalist and technological design language provides useful references for our colour matching, material processing, product labels, user guides and digital feedback styles.

In general, Hyperice shows that this smart knee pad should be positioned as a sports technology training aid, not a medical support, passive tracker or fashion-oriented fitness accessories. This direction helps to build a reliable, performance-oriented, non-clinical and beginner-friendly brand image.

12.2 Brand Analysis

After selecting Hyperice as the main brand reference, we analysed its brand language to identify the elements that can guide the final image of intelligent knee pads. Hyperice is active in the sports recovery

and performance technology market, and its product positioning is between professional sports equipment and consumer-grade health equipment. This is especially important for our products, because intelligent knee pads also need to convey technical reliability and sports support functions, and at the same time, they cannot look like clinical medical support.

Brand Identity

Hyperice conveys the brand image of high-end sports technology by focusing on recovery, sports quality and performance support. Its product design aims to present a professional and trustworthy texture, and at the same time, it is suitable for daily fitness users. This is in line with the positioning of intelligent knee support products - as a training feedback aid for gym beginners, it helps users improve their confidence and movement perception without making them feel injured or dependent on medical equipment.

Product Language

Hyperice products usually use compact appearance, simple surface, minimalist details and controlled neutral colour matching. This design language shows a sense of technology, durability and visual calmness. For intelligent knee supporters, this means that electronic modules should be visually integrated into the wearable structure, rather than being bulky or too mechanical. The product should look like a functional sports equipment, not an exposed electronic prototype.

Interaction and Feedback Language

A key feature of the Hyperice product ecosystem is the use of physical feedback, such as vibration, compression and recovery-oriented interactions. This provides a powerful reference for our tactile feedback system, in which LRA (force feedback device) and vibration motor work together to complete movement correction during squatting. Brand analysis points out that feedback should be concise, immediate and action-based, so that users can respond without interpreting complex data during exercise.

CMF and Visual Direction

Hyperice's reference design adopts dark neutral tones, matte surfaces and low-key technological details to form a unified CMF style. The dark fabric helps the product to integrate with the fitness suit and reduce the visible traces of sweat stains; the matte electronic shell highlights its durability and high-end quality. Blue or cyan as a decorative colour can be used to indicate sensor feedback, calibration and digital guidance functions. The overall design creates a consistent image of sports technology, which is both reliable and non-clinical, suitable for beginner fitness users to use easily.

13 Product Compliance

13.1 Product and User Safety

General Product Safety

AlignΞd is a consumer-oriented fitness wearable device specially designed for squat training. Therefore, it must ensure safety under normal use and reasonable and predictable use. In the United Kingdom, the 2005 General Product Safety Regulation requires that consumer goods must be safe under normal use or reasonably foreseeable conditions of use; for the EU market, Regulation (EU) 2023/988 provides a relevant General Product Safety Framework.

For *Align*Ξd products, the foreseeable risks include improper wearing, excessive compression, skin irritation, sweat contact, charging while wearing, misunderstanding of tactile prompts, and over-reliance on the product during weight training and neglecting professional guidance or coach advice. The design is designed to cope with these risks through controllable fastening range, soft and skin-friendly fabrics, rounded hard shells, mechanical rapid release devices and clear safety instructions for use.

Medical and Protective Claims Boundary

The design of *Align*Ξd aims to support safer squat training by improving movement perception and providing real-time tactile feedback. However, it is positioned as a consumer-oriented fitness training aid, not a medical device or certified personal protective equipment.

This distinction is very important, because the British Medical Device Regulation 2002 and the European Union Medical Device Regulation No. 2017/745 apply to products with medical purposes, while the Personal Protective Equipment Regulation (EU) 2016/425 is applicable to the design and manufactured to protect users from health or safety risks. The equipment made. These regulations are regarded as boundary references rather than the main compliance routes of Aligned.

Wearable Safety Implementation

As a knee-wearing product, the user safety of *Align*Ξd depends on the distribution of support during re-

peated squat flexion and extension. The protective device adopts a flexible fabric body and a soft lining, so that the pressure is evenly distributed around the thighs and calves, instead of directly concentrated above the patella or behind the knee. Rigid components (including electronic component shells and adjustment modules) are placed on the side of the protective equipment, away from the main bending area, to reduce the rigid contact caused by deep knee bending.

13.2 Electronic Regulations

Electrical Safety and Low-Voltage Electronics

The product uses a low-voltage embedded electronic system to detect the movement of the knee joint and provide tactile feedback during the movement. The main electronic components include an ESP32 microcontroller, an inertial measurement unit sensor, an ultrasonic sensor, an LRA actuator, a vibration motor and a supporting control circuit. Since these components are integrated into a wearable knee protection device, the main electrical safety risks include short circuit, overheating, line loosening caused by repeated movement of the knee joint, and direct contact with conductive components.

In order to reduce these risks, all electronic components should be isolated or wrapped from the user's skin, and a buffer device should be added to the wires around the bending area of the supporting structure. The Electrical Equipment (Safety) Ordinance of 2016 is relevant because electrical products put into the UK market must meet safety requirements before being sold. Therefore, technical documents such as circuit diagrams, component data tables, current measurements and prototype test results should be retained to support future compliance requirements.

RoHS and REACH

RoHS and REACH regulations apply because the product includes electronic components, welded joints, lines, sensors, plastic shells and wearable textile materials. RoHS regulations control the use of harmful substances used in a variety of electrical and electronic equipment, while the UK's REACH law regulations regulate the chemicals used in products marketed in the UK.

For this product, parts should be purchased from suppliers who can provide RoHS/REACH declarations as much as possible. This is especially important for those parts that are in close contact with the skin. This helps to reduce exposure to chemicals, skin irritation and long-term environmental impact.

Battery Regulations

Relevant regulations include Battery Directive No. 2006/66/EC, Waste Electronic and Electrical Equipment Directive No. 2012/19/EU, the UK Waste Battery and Storage Battery Ordinance 2009. For this wearable knee support product, these regulations mean that the battery should not be permanently sealed inside the support because it may need to be removed for replacement, repair or recycling after scrapping. The product should also display the crossed-off wheeled trash can logo to indicate that batteries and electronic components should not be disposed of with ordinary household waste. Since the product is used during exercise, the battery shell should also be easy to access safely, while avoiding the user's contact with damaged battery cells, lines or leakage risks. This helps to achieve safer maintenance, responsible handling and extend the service life of the product.

13.3 Technical Standards and Compliance Evidence

Relevant technical standards include: ISO 2768 for general manufacturing tolerances, ISO 4762 for screw fasteners, and ISO 7010 for safety signs and warning signs. For this wearable knee support device product, these standards will help ensure the unified assembly of electronic housing, battery cover and removable parts, while also improving the clarity of product labels and user instructions. In order to prove that it meets the standard, the final product needs to provide a technical document, including risk assessment, circuit diagram, material statement, battery information, component data sheet, prototype test results and user guide warning. Possible improper use should also be taken into account, such as wearing the support device too tightly, exposed to a sweaty environment, being pulled when removed, spilling water or using it with damage to the battery shell.

14 Product Labelling and User Guide Requirements

14.1 User Guide Requirements and Implementations

This user guide aims to help fitness beginners use smart knee pads safely and correctly. The content cov-

ers manual adjustment system and electronic feedback system, which is presented through concise instructions, labelling diagrams, feedback icons and safety warnings. The complete guide will be attached to the appendix and can be accessed through the QR code on the product label.

Manual Adjustment System

The guide shows how users should wrap the support around the knee and rotate the hand fastening mechanism until they feel safe and comfortable. At the same time, remind the user that if there is pain, numbness or discomfort, the product should be loosened or removed.

Electronic Calibration

Before movement, the user turns on the ESP32 Nano module and keeps it stationary, allowing the inertial measurement unit (IMU) to establish a neutral reference position. Double vibration or LED prompt to confirm the completion of calibration.

Squat Feedback

The guide explains the meaning of vibration signals. A brief LRA vibration will remind the user to move the knee inward excessively or the knee angle is unsafe, while the continuous vibration motor buzzing indicates that the target squat depth has been reached. The ultrasonic sensor also monitors the distance between the knees to support motion detection.

Charging and Care

The guide reminds users to keep electronic devices dry, remove all electronic components before cleaning, and avoid charging the product while wearing. At the same time, it is explained that the product is only a training feedback auxiliary tool, not a medical device.

14.2 Product Labelling

The product labelling system aims to convey the key compliance, disposal and maintenance information of wearable electronic knee support devices. Because the product combines fabric support and removable electronic modules, the label needs to cover electronic safety and textile maintenance requirements at the same time. The recommended labels include UKCA/CE mark position, WEEE waste disposal mark, RoHS information and textile care symbol. These labels will be placed on packaging labels, removable electronic module labels and fabric care labels to ensure that users can clearly understand the compliance, recycling requirements and correct maintenance methods of the product.

Product Compliance & Regulatory Markings

After completing the conformity assessment and product testing, the British Certification of Conduct (UKCA) and CE marks will be marked on the packaging label and removable electronic module label. These logos are important because the product contains electronic sensors, tactile actuators and battery-powered modules. In addition, basic product information, including product name, model, batch number and manufacturer's contact information, will also be displayed to support product identification, repair and traceability.



Figure 1: UKCA mark and CE mark

WEEE & Battery Recycling

Because the product contains electronic components and battery-powered removable modules, the deleted trash can symbol of waste electronic and electrical equipment (WEEE) will be marked. This reminds users that electronic modules and batteries should not be discarded with ordinary household waste, but should be disposed of through appropriate electronic waste or battery recycling channels. This label will be attached to the packaging and electronic modules to support responsible terminal disposal [4].

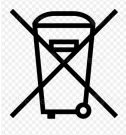


Figure 2: WEEE mark

RoHS Information

RoHS information is included because the product contains electronic components such as ESP32 mini, PCB, sensors, lines and tactile actuators. Where applicable, RoHS-compliant components will be selected to reduce the risk of harmful substances in electronic systems. Relevant information is mainly recorded in technical documents, and the packaging label is accompanied by a RoHS standard-compliant logo or statement.



Figure 3: ROHS mark

Textile Fibre Content & Care Labelling

A soft fabric care label will be attached to the inner side of the knee support to explain how to clean and maintain. Because the product is in direct contact with the skin and exposed to sweat, users need clear guidance to clean the fabric correctly so as not to damage the electronic components. The label will prompt the user to remove the removable electronic module before cleaning.

The care label will indicate "dry naturally, do not iron". The reason why natural drying is indicated is that the use of drum drying or high temperature may damage the elastic fabric, shoulder strap and electronic accessory areas. At the same time, it is marked as "cannot be ironed", because direct heating may affect the fabric structure, seams and the position of the sensor. These care instructions together help to maintain the fit, comfort and durability of the knee pad.



Figure 4: Natural drying symbol and do not iron symbol

15 Packaging Design

15.1 Packaging Requirements

Packaging needs to protect the intelligent knee support device during storage, transportation and first use. Since the product contains both soft textile knee support and removable electronic modules, the packaging must prevent damage to the fabric structure, sensor shell, tactile components and battery-powered circuits. Electronic modules should be firmly fixed to reduce movement and impact during transportation.

The packaging should also provide a clear first-time experience for gym beginners. After opening the package, users should be able to easily identify knee pads, removable electronic modules, USB-C charging cables and quick start guides. Therefore, the packaging layout should display each component in an orderly manner to help users avoid confusion and correctly understand how to wear, charge, calibrate and use the product.

15.2 Packaging Structure and Component Layout

The packaging structure is inspired by Hyperice packaging. The main products are clearly displayed, and the accessories are stored separately in special bags or compartments. This can ensure the clean experience when unpacking and prevent small parts from loosening during transportation.

For intelligent knee pads, the folded textile support will be placed in the main compartment as the main product. Detachable electronic modules, USB-C charging cables and other small electronic components will be stored together in a protected electronic bag. This can keep the electronic parts tidy, reduce the risk of loss, and prevent damage to the fabric support during transportation.

15.3 Packaging Materials and Sustainability

The outer packaging will use recyclable cardboard or corrugated paper, because it can provide sufficient protection while maintaining lightness, printability and wide recyclability. This avoids the use of plastic blister packaging and supports the brand image of the product's non-clinical sports technology.

The internal structure will be made of folding cardboard lining to fix knee supports and electronic equipment bags. This can reduce material complexity and make packaging easier to recycle. Electronic equipment bags can use recycled fabrics or paper materials to protect removable modules, USB-C data cables and small accessories, while avoiding unnecessary single-use plastic.

15.4 Production and Manufacturing Process

The outer box will be made of cardboard through die-cutting, indentation, folding and bonding. The die-cutting is used to form the outline, sealing strip and opening of the box, while the indentation can produce clear folds, which is convenient for precise assembly. The process is simple, repeatable, and suitable for recyclable cardboard packaging.

The internal folded cardboard lining will be made with the same die-cutting and indentation process. By folding, multiple compartments can be formed, which can firmly fix knee pads and electronic device storage bags without the need to use plastic trays. This makes the packaging lighter, easier to assemble, and easy to recycle. In addition, the patterns on the packaging will be prototyped by digital printing, or flexographic printing will be used in mass production.

15.5 Graphic Design

The packaging graphic design takes reference from the Hypervolt 2 Pro packaging, particularly its clean sports-tech layout, strong product imagery and minimal information hierarchy. The following features would be adapted for the smart knee support packaging:

Large product render: the front of the box would use a clear image of the smart knee support as the main visual focus. Minimal front information: only the product name, core function and key feature icons would be shown to avoid visual clutter. High-contrast layout: dark and light areas would be used to create a premium, technical appearance and improve readability. Simple icon system: key functions such as knee alignment feedback, squat-depth guidance and detachable electronics would be communicated through concise icons. Consistent typography: the same font style and visual hierarchy would be used across the packaging, product label and user guide. Organised brand presentation: the packaging would feel clean, structured and performance-focused, aligning with the product's sports-tech identity.

Overall, this graphic approach keeps the packaging clear and professional while maintaining consistency across the product, documentation and final brand presentation.

16 User Journey and Experience

AlignΞd began with users at the centre, a want to help people, specifically those who are underserved, and it led to this monumental product. But what will allow the product and brand to stay true is to keep users at the centre, just as they began. This can be seen in the deliberate design decisions that uplift the user experience throughout the journey of use of the product.

16.1 User Experience

As stated prior, each decision with regards to the product was made with users and their experience in mind. This can be seen through, but not limited to, the following:

Magnetic Connection and Guide Pins The connection between the dock plate and electronics is both satisfying and fool-proof. The magnets provide a crisp connection and the opposing polarities and guide pins only allow the user to put it in the correct direction, adding to the positive user experience.

TPU outer casing A deliberate choice was made to user two shot moulding to allow the outer shell to be made of TPU, providing a sleek and elegant hand-feel.

Detachable Pod Arguably the strongest UX feature of the product, the detachable electronics pod provides ease and accessibility from two fronts: allowing the user to easy charge the device, and allowing them to safely and simply be washed.

Patella Cutout Although it may seem like a simple addition, this features serves the user on many fronts. First, it reduces tension on the patella during bending while still supporting the rest of the knee, significantly reducing the risk of injury. As well as this, it makes the correct orientation of the brace clear to the user, reducing any chance of setup error.

17 Final Design Summary

Our final product consists of 4 main parts: The knee brace itself, which is a standard available knee brace, as we were not aiming to change the biomechanics of them, but in this we have housed our vibration motors. These are wired to a baseplate.

Connecting to this base is the electronics pod, containing the core of the brace's functionality. This pod has been designed with asymmetrical pegs and magnets to ensure the user can only place the pod on correctly, and secures itself easily. It can be charged with a USB C cable, a widely used and European-standardised connector. The core contains an ESP32 mini, as we did not need a lot of computing power, but needed to house the electronics in a small space. An IMU has been used for the positional tracking to determine correct/incorrect knee valgus, which determines which motors vibrate and by how much, to indicate to the user to move their knees to correct their form.

When produced in industry, the lid of the pod will be ultrasonically welded to the body, keeping it fixed securely and acting as a water-tight barrier, crucial for use in the gym where people sweat and may spill water as they drink.

On the other side of the brace is our BOA-inspired tightening mechanism, widely used within technical shoes, where the user can easily tighten or loosen the brace with one hand. This was implemented to make it easier to use the brace.

18 Analysis of Prototyping and Testing

Whilst aiming to follow our schedule, we opted to make adjustments to optimise certain processes. Ordering a PCB would have taken too long to arrive, and

so we used a perforated board to solder components to instead, and this allowed us to still reduce the size compared to a breadboard.

The mechanism went through numerous iterations- we experimented with worm gears but needed a way to release the tension, to reduce strain on the motor, rather than constantly powering it. This eventually resulted in us adapting a recreation of a BOA system as this solved our issue.

The electronics functionally was working well early on, and the code was written well. The IMU had no drift when measuring the knee angle, and calibrated itself each time, which was great as we could trust it to detect correct/incorrect knee position. The only setbacks came later during soldering, however functionally the code worked very well. In industry we would opt to use a PCB instead, as they are mass producible (can print multiple on a single sheet), they take up less space and is much faster than soldering by hand.

Ideally, we would have tested our product more in-use than functionally in a lab, and with potential users for validation however due to time constraints this was not possible.

19 Further Developments

As with any good product, a company must look to the future in terms of growth and further improvement. This comes in two forms: response to customer feedback on our product, and innovation from the engineers and designers. In this case, the plans are as follows:

19.1 Customer Outreach

One of the best ways to iterate and improve, is to gather user feedback. Although we do our best to predict what users need and want, it typically does not always reflect reality, and their suggestions can come as a surprise. As such, during the launch phase of our first knee brace, we would implore users to give us their views on a number of aspects of the product:

UI: Is it truly easy to use and simple to understand? Does the product make them feel out of place at the gym/do they think it draws extra attention to them? One side to be wary of here is perceptual expectancy, like in the Dartmouth scar experiment, however this feedback will still prove insightful.

Physical feedback: Do users feel that the physical feedback is enough, or would pairing it with a digital output be more welcoming to them? Although functionally not necessary, would having another brace on the other leg feel more natural than only having it on one and receiving vibrations on only one leg?

19.2 Internal Innovation

App: We opted not to create an app at this stage, however it is definitely something to consider down the pipeline, as it would allow users to track trends such as how often they have incorrect positions, but perhaps more valuable, it could record their sets and reps, which is what every gym-goer keeps a track of, so they remember what weights they can safely push to.

Physical Feedback: Although we explored self-tightening, at the time it seemed redundant, as it was not designed as an accessibility device. However, we could implement a dynamic tightening system, where the brace tightens or loosens, and applies pressure to different regions of you legs based on your squat position. This would feel like someone physically correcting your position, rather than a suggestive indicator i.e vibrations. We could accomplish this using pneumatics, with areas of the brace divided into sections that can be pressurised, much like blood pressure monitors do but at specific locations.

Other exercises: Our product can be adapted to be used in other exercises by using it on different parts of the body. For example, if placed on an arm, dumbbell bench press could be monitored and corrected as there is a lot of unnecessary movement associated with it in beginners, due to incorrect form and weak arms. Different braces may need to be produced, as if they were being wrapped around different muscle groups, they will fit differently. However, the core electronics pod should be unchanged, and could be updated to work on any new exercises added to Hyperice X AlignΞd product list. This would therefore make the initial purchase the bigger one, with subsequent ones primarily for the brace and sensors situated within.

20 Project Planning

20.1 Project Timeline

The main structure of the Gantt chart remained consistent, however a few design decisions resulted

in some time frame changes and the addition of a set time for external production and delivery time. Initially, we splpit our work into four phases:

Phase 1: Test IMU angle sensing. Use IR. Test LRA and if you can feel it through material (how much power it needs). Test vibration motor and if you can feel it through material (how much power it needs) Mechanical, hand-pulled worm gear system to test if tightening works and if it is possible. Create a lo-fi general shape and material of the knee brace.

Phase 2: Test sensing subsystem together. IMU + LRA + Vibration, how they work together, needed power output. Using a motor + gears to tighten, needed gear ratio + motor torque to tighten in under x seconds (requirement),power needed. Material testing(box).

Phase 3: Whole system working together. Encasing sizing, shape, material. Decide battery requirements.

Phase 4: Refine final product size, shape, and optimise each component in the whole system.

20.2 Unplanned Changes

Setbacks during the mechanism assembly and electronics soldering resulted in our later milestones being pushed back, leaving us with less time for finishing touches. As we made compromises based on combining the sub assemblies, some renders of the pod and final product needed to be redone, as they were no longer consistent with the physical prototype.

The original schedule assumed the production parts (ICM-20948 + DRV2605L/LRA), and ardruino nano ESP32; in practice we substituted for a QT Py ESP32-S2 as it was much smaller, with a BNO055 for motion due to us frying the original IMU.

20.3 Analysis

The term started off at a good pace, but slowed down as we encountered issues during our assembly stage. Zayn and William focused on the electronics, whilst Yousuf and James worked on the mechanism. We opted to split up the processes, so that we could complete the sub-systems simultaneously and have ample time for putting them together.

Zayn primarily wrote the technical report. Yousuf, James and William worked on the portfolio, with James and William allocated the rendering and visual tasks. Yousuf and Zayn created the CAD models used, and Yousuf laid out the portfolio. James created the submission video of our project. Zayn made the presentation slides and directed the composition of our project.

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22 Appendix

22.1 Complete Bill of Materials

#	Component	Qty	Material / part
1	Lower shell	1	PA6-GF30
2	Upper lid (2K)	1	PA6-GF30 + TPU 95A
3	Dock plate	1	PA6-GF30
4	FR4 contact sub-plate	1	FR4 / ENIG
5	Pogo pins	4	Spring probe, Au
6	Alignment magnets	4	N52 NdFeB 8×3
7	PCB assembly	1	FR4 2-layer, ENIG, all-SMD
8	Microcontroller	1	Arduino Nano ESP32 (ESP32-S3)
9	IMU	1	TDK ICM-20948
10	Haptic driver	1	TI DRV2605L
11	LRA (valgus tap)	1	10 mm coin, 175 Hz
12	ERM (depth buzz)	1	10 mm coin, 3 V
13	Battery	1	LiPo 3.7 V 300 mAh
14	FPC interconnect	1	0.2 mm polyimide, 4-cond.
15	Actuator pockets	2	LSR 40 Shore A (ISO 10993)
16	Neoprene sleeve	1	SCR neoprene 3 mm, jersey
17	Patella ring	1	TPU
18	Cord guide clips	6	PA6-GF15
19	Boa dial + cord	1	Licensed; Dyneema SK75 1.5 mm
20	USB-C / charge circuit	1	On-PCB

22.2 IMU Test Code

22.3 Electronics SubSystem Code

```
#include <Wire.h>
#include <Adafruit_ICM20X.h>
#include <Adafruit_ICM20948.h>
#include <Adafruit_Sensor.h>
#include <Adafruit_DRV2605.h>

// ============================================================
// Smart Knee Brace Prototype
// Hardware:
// - Arduino Nano ESP32
// - ICM20948 IMU sensor
// - DRV2605L haptic driver + LRA motor
// - ERM coin vibration motor controlled through 2N7000 MOSFET
```

```
//
// Function:
// - IMU estimates knee/leg pitch and roll angle
// - LRA gives a strong shock/click when knee valgus is detected
// - ERM vibration motor gives a longer buzz when squat depth is good
// ============================================================

Adafruit_ICM20948 icm;
Adafruit_DRV2605 drv;

// ------------------------------------------------------------
// ERM vibration motor pin
// D5 goes to 100 ohm resistor -> 2N7000 Gate
// ------------------------------------------------------------
const int ERM_PIN = D5;

// ------------------------------------------------------------
// Thresholds     tune these during testing
// ------------------------------------------------------------
const float VALGUS_THRESHOLD  = 8.0;    // roll angle in degrees
const float SQUAT_THRESHOLD   = -30.0;  // pitch angle in degrees
const float RETURN_HYSTERESIS = 2.0;    // prevents flickering

// ------------------------------------------------------------
// LRA haptic effect through DRV2605L
// ------------------------------------------------------------
const uint8_t EFFECT_VALGUS = 1;        // strong click / shock feedback

// ------------------------------------------------------------
// ERM motor vibration strength
// analogWrite range: 0 to 255
// ------------------------------------------------------------
const int ERM_DEPTH_POWER = 204;        // about 80% power
const int ERM_DEPTH_TIME  = 900;        // longer buzz for good depth

// ------------------------------------------------------------
// Complementary filter
// ------------------------------------------------------------
float compPitch = 0.0;
float compRoll  = 0.0;

unsigned long lastTime = 0;
const float ALPHA = 0.96;

// ------------------------------------------------------------
// Gyro calibration
// ------------------------------------------------------------
float gx_bias = 0.0;
float gy_bias = 0.0;
float gz_bias = 0.0;

const int CAL_SAMPLES = 500;

// ------------------------------------------------------------
// Detection state
// ------------------------------------------------------------
bool inValgus = false;
bool atDepth  = false;

// ------------------------------------------------------------
// Cooldowns to stop repeated vibration
```

```
// ------------------------------------------------------------
unsigned long lastLRAFire = 0;
unsigned long lastERMFire = 0;

const unsigned long LRA_COOLDOWN = 600;   // ms
const unsigned long ERM_COOLDOWN = 1200;  // ms

// ============================================================
// Gyro calibration
// ============================================================
void calibrateGyro() {
  Serial.println("Calibrating gyro     keep the IMU still...");
  delay(2000);

  double sx = 0.0;
  double sy = 0.0;
  double sz = 0.0;

  for (int i = 0; i < CAL_SAMPLES; i++) {
    sensors_event_t accel;
    sensors_event_t gyro;
    sensors_event_t temp;
    sensors_event_t mag;

    icm.getEvent(&accel, &gyro, &temp, &mag);

    sx += gyro.gyro.x;
    sy += gyro.gyro.y;
    sz += gyro.gyro.z;

    delay(4);
  }

  gx_bias = sx / CAL_SAMPLES;
  gy_bias = sy / CAL_SAMPLES;
  gz_bias = sz / CAL_SAMPLES;

  Serial.println("Gyro calibration complete.");
}

// ============================================================
// LRA calibration for DRV2605L
// ============================================================
void calibrateLRA() {
  Serial.println("Calibrating LRA     keep motor still...");

  // Auto-calibration mode
  drv.writeRegister8(0x01, 0x07);

  // Rated voltage and overdrive clamp for small 3V LRA
  drv.writeRegister8(0x16, 0x3E); // RATED_VOLTAGE
  drv.writeRegister8(0x17, 0x57); // OD_CLAMP

  // LRA mode settings
  drv.writeRegister8(0x1A, 0xB6); // FEEDBACK_CONTROL: LRA mode
  drv.writeRegister8(0x1B, 0x93); // CONTROL1: startup boost
  drv.writeRegister8(0x1C, 0xF5); // CONTROL2
  drv.writeRegister8(0x1D, 0xA0); // CONTROL3

  // Start calibration
  drv.writeRegister8(0x0C, 0x01);
```

```
136
137   unsigned long start = millis();
138
139   while (drv.readRegister8(0x0C) & 0x01) {
140     delay(25);
141
142     if (millis() - start > 2000) {
143       Serial.println("LRA calibration timed out.");
144       return;
145     }
146   }
147
148   uint8_t status = drv.readRegister8(0x00);
149
150   if (status & 0x08) {
151     Serial.println("LRA calibration FAILED     check LRA wiring.");
152   } else {
153     Serial.println("LRA calibration OK.");
154   }
155
156   // Return to internal trigger mode
157   drv.writeRegister8(0x01, 0x00);
158   drv.selectLibrary(6);
159 }
160
161 // ============================================================
162 // Fire LRA strong shock/click
163 // ============================================================
164 void fireLRA(uint8_t effect) {
165   unsigned long now = millis();
166
167   if (now - lastLRAFire < LRA_COOLDOWN) {
168     return;
169   }
170
171   lastLRAFire = now;
172
173   drv.setWaveform(0, effect);
174   drv.setWaveform(1, 0);
175   drv.go();
176
177   Serial.println("LRA shock fired.");
178 }
179
180 // ============================================================
181 // Fire ERM long buzz
182 // ============================================================
183 void fireERMLongBuzz() {
184   unsigned long now = millis();
185
186   if (now - lastERMFire < ERM_COOLDOWN) {
187     return;
188   }
189
190   lastERMFire = now;
191
192   Serial.println("ERM long buzz fired - good squat depth.");
193
194   analogWrite(ERM_PIN, ERM_DEPTH_POWER);
195   delay(ERM_DEPTH_TIME);
196   analogWrite(ERM_PIN, 0);
```

```
}

// ============================================================
// Setup
// ============================================================
void setup() {
  Serial.begin(115200);
  while (!Serial) {
    delay(10);
  }

  Wire.begin();

  pinMode(ERM_PIN, OUTPUT);
  analogWrite(ERM_PIN, 0);

  Serial.println("Smart Knee Brace starting...");

  // ------------------------------------------------------------
  // Initialise IMU
  // ------------------------------------------------------------
  if (!icm.begin_I2C()) {
    Serial.println("ICM20948 not found     check SDA/SCL wiring.");
    while (1) {
      delay(10);
    }
  }

  icm.setAccelRange(ICM20948_ACCEL_RANGE_4_G);
  icm.setGyroRange(ICM20948_GYRO_RANGE_500_DPS);

  Serial.println("ICM20948 IMU found.");

  // ------------------------------------------------------------
  // Initialise DRV2605L
  // ------------------------------------------------------------
  if (!drv.begin()) {
    Serial.println("DRV2605L not found     check SDA/SCL wiring.");
    while (1) {
      delay(10);
    }
  }

  Serial.println("DRV2605L found.");

  // ------------------------------------------------------------
  // Calibrate sensors
  // ------------------------------------------------------------
  calibrateGyro();
  calibrateLRA();

  // ------------------------------------------------------------
  // Seed complementary filter from accelerometer angle
  // ------------------------------------------------------------
  sensors_event_t accel;
  sensors_event_t gyro;
  sensors_event_t temp;
  sensors_event_t mag;

  icm.getEvent(&accel, &gyro, &temp, &mag);

```

```
  compPitch = atan2(
                accel.acceleration.y,
                sqrt(accel.acceleration.x * accel.acceleration.x +
                     accel.acceleration.z * accel.acceleration.z)
              ) * 180.0 / PI;

  compRoll = atan2(
               -accel.acceleration.x,
               accel.acceleration.z
             ) * 180.0 / PI;

  lastTime = micros();

  Serial.println("READY");
  Serial.println("CSV format:");
  Serial.println("millis,pitch,roll,valgus_flag,depth_flag");
}

// ============================================================
// Main loop
// ============================================================
void loop() {
  sensors_event_t accel;
  sensors_event_t gyro;
  sensors_event_t temp;
  sensors_event_t mag;

  icm.getEvent(&accel, &gyro, &temp, &mag);

  unsigned long nowMicros = micros();
  float dt = (nowMicros - lastTime) / 1000000.0f;
  lastTime = nowMicros;

  // ------------------------------------------------------------
  // Remove gyro bias
  // Gyro units are rad/s, so convert to degrees/s later
  // ------------------------------------------------------------
  float gx = gyro.gyro.x - gx_bias;
  float gy = gyro.gyro.y - gy_bias;

  // ------------------------------------------------------------
  // Accelerometer angle estimate
  // ------------------------------------------------------------
  float accelPitch = atan2(
                       accel.acceleration.y,
                       sqrt(accel.acceleration.x * accel.acceleration.x +
                            accel.acceleration.z * accel.acceleration.z)
                     ) * 180.0 / PI;

  float accelRoll = atan2(
                      -accel.acceleration.x,
                      accel.acceleration.z
                    ) * 180.0 / PI;

  // ------------------------------------------------------------
  // Complementary filter
  // ------------------------------------------------------------
  compPitch = ALPHA * (compPitch + gy * dt * 180.0 / PI)
              + (1.0 - ALPHA) * accelPitch;

  compRoll = ALPHA * (compRoll + gx * dt * 180.0 / PI)
```

```
            + (1.0 - ALPHA) * accelRoll;

  // ---------------------------------------------------------
  // Valgus detection
  // If roll is too large, LRA gives strong shock/click
  // ---------------------------------------------------------
  float absRoll = abs(compRoll);

  if (!inValgus && absRoll > VALGUS_THRESHOLD) {
    inValgus = true;
    fireLRA(EFFECT_VALGUS);
  }
  else if (inValgus && absRoll < VALGUS_THRESHOLD - RETURN_HYSTERESIS) {
    inValgus = false;
  }

  // ---------------------------------------------------------
  // Squat depth detection
  // If pitch reaches good depth, ERM gives longer buzz
  // ---------------------------------------------------------
  if (!atDepth && compPitch < SQUAT_THRESHOLD) {
    atDepth = true;
    fireERMLongBuzz();
  }
  else if (atDepth && compPitch > SQUAT_THRESHOLD + RETURN_HYSTERESIS) {
    atDepth = false;
  }

  // ---------------------------------------------------------
  // Serial output for plotting / debugging
  // ---------------------------------------------------------
  Serial.print(millis());
  Serial.print(",");
  Serial.print(compPitch, 2);
  Serial.print(",");
  Serial.print(compRoll, 2);
  Serial.print(",");
  Serial.print(inValgus ? 1 : 0);
  Serial.print(",");
  Serial.println(atDepth ? 1 : 0);

  delay(10); // about 100 Hz
}
```

Listing 1: Smart Knee Brace Prototype Arduino Code